

Through the Looking Cube: Solving the Rubik's Cube from Within

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Figure 1: The player manipulates a physical Rubik's Cube while seeing it from the inside through a VR headset. The cube operations affect the virtual environment accordingly.

Abstract

We present an interactive VR experience where users solve a physical Rubik's Cube while seeing it from the inside. This unusual game mechanic challenges natural spatial perception mechanisms, leading to ambiguity regarding their actual position with respect to the object in their hands. Beyond puzzle-solving, we applied this interaction paradigm to 3D scenes that must be reconstructed through Rubik's Cube operations in ways that some users described as akin to "being inside a work of Escher." We discuss the technical implementations, document participant interactions, and discuss

implications for artistic expression, game design, and embodiment in VR settings.

CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → **Virtual reality**.

Keywords

New Media Art, Virtual Reality, Embodiment, Immersion, Spatial Perception, Game Design

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1 Introduction

Many action video games challenge players to navigate complex 2D and 3D environments, including interiors or open spaces. Leaving

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aside fantastic content, the geometry of the level could be embedded in the real world, and the same can be said for most puzzle video games. Although the complexity of the space may put a strain on spatial skills and memory, these games present no ambiguity regarding our position in the scenes or the geometry of objects: players just see themselves as avatars (third person view), or observe the scene as if they were immersed in it (first person view). On the other hand, a special niche exists in both categories for games that *have* the player navigate spaces or deal with objects that could *not* live in the physical world. The environments may contain extra dimensions [Kortemeyer et al. 2013], wormholes or “portals” connecting distant places [Valve Corporation 2007], or deal with forced perspective and “impossible” objects [Bruce 2013]. These “paradoxical spaces,” reminiscent (and perhaps inspired) by the works of Escher, force us to dispense with our natural understanding of space, or the consequences of our own physical actions. In particular, the certainty of *position and point of view*, and such concepts as *inside or outside of things* become non-operational and fuzzy, and relying on them actually hinders performance. Thanks to the capacity of Virtual Reality to fully immerse the user in a simulated environment, games such as Superliminal [Pillow Castle Games 2019], bring disorientation to a whole new level.

In this paper, we present a VR-based experimental artwork inspired by the mechanics of the classical Rubik’s Cube 3D puzzle. It employs the physical cube as an interface to deconstruct 3D virtual scenes with an additional twist: players are physically outside, but virtually inside the cube they manipulate in their hands (1).

Solving the real Rubik’s Cube recruits a battery of cognitive abilities ranging from spatial memory to keep track of the colors, intuitive understanding of space and the capacity to predict the effects of spatial transformations through hand-eye coordination. As will be described below, actually solving the cube is almost impossible - it is a pretext to and a mean to heightens the appreciation of this paradoxical experience.

2 Related Works

The representation of space is a fundamental subject in art. The development of perspective in the Renaissance allowed for realistic representation of reality, but perhaps more importantly, understanding the rules helped deconstruct them in a systematic way, thus opening novel opportunities for experimentation. This artistic practice worked synergistically with scientific inquiry into the mechanisms of human perception. Relevant to our work is the question of how we situate ourselves with respect to the surrounding and how we define the inside and the outside of spaces. The certainty of being present *somewhere* seems to carry an absolute truth. *I see, therefore I am somewhere* - regardless of whether in a virtual or real space [Slater and Sanchez-Vives 2016]. However, artists and scientists have shown us time and again that the sense of placement and presence is a construct of the mind that can be easily manipulated.

2.1 Paradoxical spaces in art

The well-known work of M.C. Escher explores paradoxical spaces and impossible geometries through intricate drawings and prints. Iconic pieces such as “Ascending and Descending” and “Relativity”

challenge our conception of space, while others such as “Drawing Hands” trap the viewer into infinitely recursive spaces, making it impossible to know where they really stand as an observer. By exploiting mathematical concepts and visual illusions, Escher’s work demonstrates how our perception of space can be thwarted, resulting in compositions that simultaneously appear logical yet defy physical reality (Escher was a friend of the great physicist Roger Penrose, also fond of impossible figures and inventor of the famous “Penrose triangle”) [Hollist 2000]. In the same vein, in Escher’s art, foreground and background often become interchangeable, generating contradictory but complementary interpretations of the work.

2.2 Paradoxical spaces and Video Games

Exploiting paradoxical spaces is not uncommon in video games. Portal [Valve Corporation 2007] popularized this type of gameplay by allowing players to create and move through portals interconnecting distant places, while Monument Valley [ustwo games 2014] drew direct inspiration from Escher’s impossible architectures to create puzzle mechanics based on perspective manipulation. Antichamber [Bruce 2013] further pushed the boundaries by implementing non-Euclidean geometry and recursive environments containing themselves.

Games like Manifold Garden [William Chyr Studio 2019] and Echochrome [Japan Studio 2008] demonstrate how paradoxical spaces and an appropriate game mechanic can force players to relearn their intuitive understanding of spatial relationships to progress.

In terms of interaction, LevelHead [Oliver 2007] is an experimental augmented reality game taking place on the faces of a physical cube. Using AR gear, the player can see scrambled parts of a building through its faces; by tilting the cube, it is possible to guide a virtual character toward an exit. The use of the physical cube as the interface makes it particularly relevant to our work as will be described below.

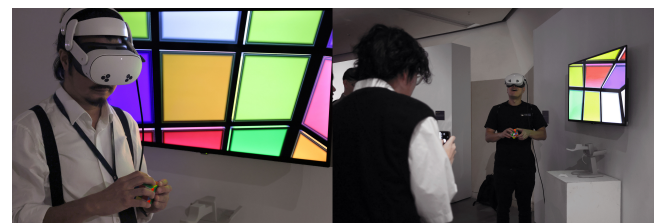


Figure 2: Setup of Through the Looking Cube in Exhibition. The TV on the wall displays the real-time images transmitted from the VR headset through the first-person view.

3 Solving the Rubik’s Cube from Within: the experience

Upon wearing the headset, players find themselves in what appears to be the interior of a room-sized Rubik’s cube. As users operate on the physical version they hold in their hands, the giant walls around them rotate accordingly (2).

While solving the real Rubik's Cube, players must rely on hand-eye coordination, continuously rotating the object to search for facets. To do the same in the VR setting, players must instead turn their heads or whole bodies to scan the surrounding walls.

The virtual environment maintains perfect synchronization with the physical cube, creating a compelling sense of being inside the physical puzzle they are attempting to solve. This inside-outside ambivalence generates an interesting cognitive dissonance that is enjoyable and new, regardless of the advancement toward the goal.

4 Technical Details

4.1 Custom Unity-based VR environment

The VR environment is built with Steam VR and the Unity engine [tourist7r 2018]. We employed an Oculus Quest 3S VR headset linked wirelessly to a Windows laptop PC with the support of the Meta Quest Link [Meta 2025]. The VR hand controllers are disabled as the physical Rubik's Cube acts as the controller 1.

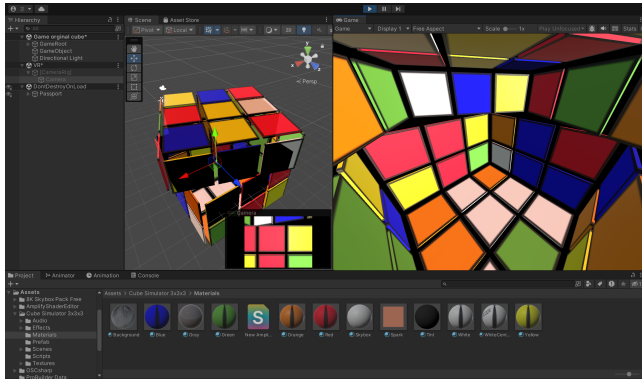


Figure 3: Screenshot in Unity Editor, 3D models in editor's view (left), Player's view in VR (right)

At present, the global orientation of the virtual cube remains unchanged with respect to the real space. This is a technical limitation as the cube does not integrate inertial sensors and, as will be described below, can have consequences on the way participants come up with manipulation strategies.

4.2 Tracking of physical moves

To track the physical moves, we hacked the QiYi QYSC Bluetooth Smart Cube [QiYiCube [n. d.]]. It is designed to log time scores or for training, and can run an automatic solver [cs0x7f [n. d.]]. Reverse engineering the Bluetooth protocol was facilitated by open-source software [Flying-Toast 2024]. A custom JavaScript local server [nodejs 2025] works as the middle tier handling the Bluetooth connection, decoding the data packet, and informing the Unity server through custom OSC messages (4). Interested readers can find the relevant code in our webpage [Bai et al. 2025].

Given the unexpected difficulty of solving the cube from the inside, we removed time constraints. Originally, the passing of time was accompanied by a gradual reduction of the cube's size, squeezing the user and ending the game.

5 Beyond gameplay: Rubik's Cubism

Throughout history, architecture, art, and cinema have employed techniques like forced perspective, anamorphosis, and optical illusions to evoke awe, convey images of power, and situate the individual within a grand narrative. Gothic cathedrals, for example, use towering spires and vast interiors to dwarf the human scale, drawing the eye upward toward the heavens and symbolizing divine transcendence.

In the early XX century, artists engaged in an opposite, more introspective direction through the fragmentation of perception, confining or replacing the subject with abstract and geometric shapes and challenging the passive role of the observer and making viewpoints relative. Precisely for this reason, experimental video games can learn from Cubist or Modernist experimentation.

We also experimented in this direction with our setup by replacing the colored facets by transparent glass panels with a view over a large 3D scene - see fig (5). However, the players initially encounter these panels scrambled around the walls; to solve this three-dimensional jigsaw, the players must operate on the physical Rubik's Cube. Interestingly, their position inside the cube and the orientation of the window affect parallax, so the views through the same panel are not exactly the same. Because of the added complexity, solving the puzzle becomes a secondary goal and can induce introspective thoughts as to how our mind works. As Georges Braque elaborates, "The senses distort, the mind forms" [Fry 1966]. Again, the experience clearly evokes Cubist experiments, albeit with an interactive twist and fully immersed in virtual reality.

6 Informal observations and Discussion

Through the Looking Cube was displayed at the 19th SIG playful media showcase at the School of Creative Media. Even if the following account is a far cry from a serious and controlled user study, two weeks of informal interaction with the public prompted the writing of this paper as we felt that the experience could be of interest to artists, game designers, or prompt researchers in virtual reality or cognitive sciences to actually develop larger scale and controlled experiments.

6.1 Mitigating spatial disorientation through the reduction of sensory input

"In VR, I cannot see and touch the Rubik's Cube as usual. Every time I stop, I need to reconstruct a cube model in my mind out of thin air. Could be brain-burning." [translated from Mandarin]

We conducted informal interviews with several participants. One of the interviewees with experience solving the cube went through the "recipe formulas" quickly. However, when only the top surface was left unassembled, he paradoxically lost interest in solving it, explaining that the additional process of re-imagining consumed the majority of his mental efforts, and that he experienced stress from having to reconfigure the cube each time he proceeded with the systematic operations. Interestingly, when applying the mechanical sequence, he would block the interfering, immersive visual experience, perhaps the cause of his anxiety.

In contrast, novices spent more time freely exploring the environment and interaction mechanisms, and showed more signs of enjoyment. This is perhaps not surprising, as novices ignore

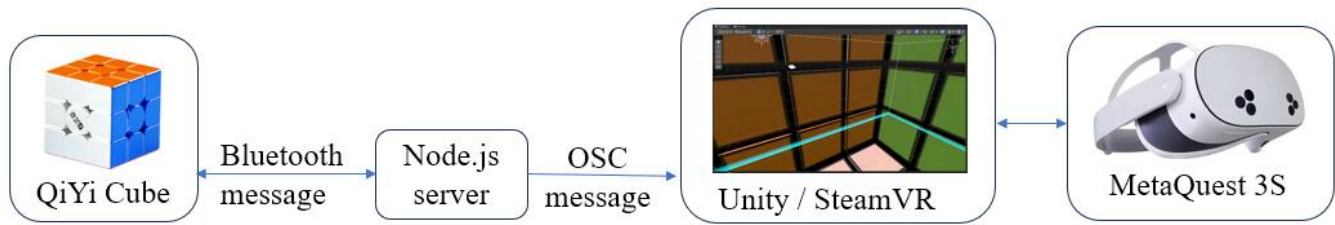


Figure 4: Communication among Bluetooth Cube, Unity and VR device

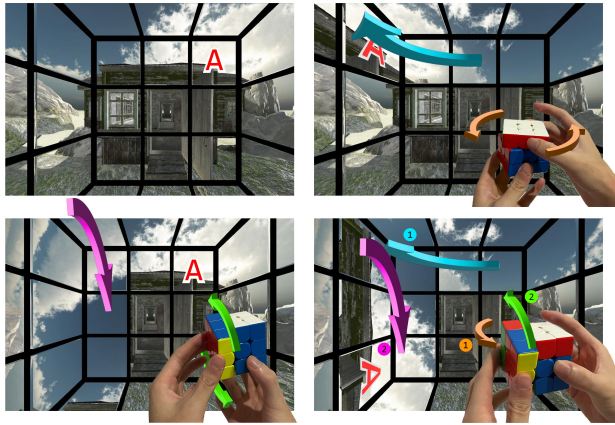


Figure 5: The window pane A affected by a 90 degrees clockwise rotation of the top layer of the cube. Note that the view through the window A remain the same if the user rotates the left layer instead. The image on the bottom left shows the final position of the window pane when both operations are executed sequentially. Although difficult to recognize in the figure, the unchanged position of the player inside the cube affect parallax

“formulas” and probably treat the real Rubik’s Cube as a jigsaw, searching and moving pieces around in a very inefficient yet enjoyable and mind-absorbing way. Although they never fully solved the cube -at best one or two faces - completing a face prompted a more evident expression of joy and relief: perhaps one or two solid walls clear some ambiguity regarding the relative position and orientation in the virtual space, thus reducing disorientation and anxiety.

6.2 Coping with self-location ambiguity and lack of direct hand-eye coordination

“When I turn the left side of the Rubik’s Cube in my hand, the face I see on the left stays static, but the face on the top moves. Sometimes I get confused if I try to think about the relationship between them.” [translated from Mandarin]

As Merleau-Ponty argues, space is not a “Newtonian, objective or uniform container”, but a schema of the body [Merleau-Ponty 1962]. The ability to move and the intentionality of the body define the perception of depth, direction and distance. Today, these

views are synthesized in Alva Noë’s “Action in Perception” theory, positing the dependence of what we call sensorial modalities, including the sense of “presence”, on the means of interaction with the world. Take this even further, suggesting that perception is not just influenced by actions, but is itself a kind of skillful activity [Noë 2004].

Location ambiguity and lack of direct hand-eye coordination in *Through the Looking Cube* challenges our natural sense of physical embodiment and presence. VR-triggered “out-of-body” experiments are well documented. Cognitive science researcher Olaf Blanke used VR technology to study the phantom sense of body ownership [Blanke 2012]. Artwork *Boxed Ego* [Cassinelli and Ishikawa 2008] creates a recursive experience where observers peer inside a box to discover a 3D miniature version of themselves, peering inside an even smaller box and creating a fleeting sense of disembodiment in real space. Video game Patrick’s Parabox [Traynor 2022] is also keen on infinite recursion puzzles and block spaces. Similarly, *Through the Looking Cube* generates ambiguity about the location of the self. While participants know - and feel - they are handling a small physical cube in their hands, they perceive it from the inside, creating the eerie feeling that they are somehow handling a miniature version of themselves.

6.3 On the nature of space and the sense of presence

“You have to match in your mind what you see inside with what you should see outside.” [translated from Mandarin]

From the point of view of cognitive sciences, the project raises broader questions regarding how the mind constructs space and infers location and shape of things by integrating proprioceptive and tactile information. Indeed, our experiment breaks hand-eye coordination and decouples visual and tactile information.

Although a physicist, Henri Poincaré shared some of Merleau-Ponty views: aspects of space don’t exist on their own, but rather are something our minds create based on how we move and interact with the world [Poincaré 1902]. Views like these are synthesized in Alva Noë’s “Action in Perception”, while taking them further by suggesting that perception is not just tied to action, but is itself a kind of skillful *activity* [Noë 2004].

6.4 Virtual embodiment eliminating distinction between interior and exterior space

“It reminds me that some users scroll the mouse wheel down to turn down the page, others scroll the opposite way; it all depends on

how you consider the controller. If we just treat the Rubik's Cube in our hands as a controller, without imagining what it looks like, but instead focus on looking around, it will become much simpler." [translated from Mandarin]

In line with Merleau-Ponty, the theory of Embodied Cognition (and Embodied Mind) posits that the physical body shapes how we make sense of the world and even how we think [Varela et al. 2017]. The constant shifting between interior and exterior perspectives creates a controlled form of spatial disorientation. This dual role creates a unique phenomenological experience where the boundaries between subject and object become fluid. Here, there is no pure interior or exterior space, but oriented spaces centered on the body.

As described before, we observed that when confronted with the necessity to actually *solve* the puzzle, participants seem to unconsciously gravitate toward the best operational interpretation of the situation: inside or outside. Because of their reflex to engage in gameplay rather than freely explore this novel experience, they may not feel themselves simultaneously "inside and outside" as much as novices, and they rely on visual cues only when necessary to update the state of the cube. On the other hand, novices engage more openly with the immersive experience, feeling themselves inside the device and perhaps treating the physical cube simply as an interface for spatial navigation. Perhaps depending on the task at hand, these interpretations may flip like a Necker Cube in our minds.

7 A possible source of inspiration for artistic expression?

On the basis of eliminating interior and exterior, we could further explore how humans create a mind map of the space they are in when they cannot see the full appearance. While manipulating the cube, we also physically orient ourselves in space, thereby forming a profound intuitive understanding of the directional concepts inside and outside the block.

In this framework, paradoxical spaces like those depicted in Escher's paintings can be built in order to explore the effects of spatial illusions on cognition or become experimental material for spatial reconstruction exercises. If one side of the Rubik's Cube needs two full turns to return to its original state, like a fermion in quantum physics, how would players map their body potential in such space? And how do they embody in a world with different rules? In the spatial structure different from the real world, the game mechanics and objectives can be redesigned to achieve an experience that is both playable and valuable for research regarding cognition and embodiment.

As described above, some described the experience as psychedelic and enjoyed the contemplation of the broken virtual space rather than concentrating on solving the cube. This reminded us of our attempt to map the scene to six faces of the Cube (5). In VR perspective, while significantly increasing the difficulty of solving the cube, the player can also stop and appreciate mixed images on disrupted faces.

We notice that visual disturbance or fragmentation is likely to bring a brand-new artistic experience. Monet suffered from

cataracts in his later years, which probably reduced his yellow vision and edge sensitivity to a considerable extent, and great works such as *Waterlilies* were created under completely different visual perception [Marmor 2006]. In this regard, optical neurological diseases also provide inspiration. Käthe Kollwitz is inferred to have Alice in Wonderland syndrome (AIWS) - a disease caused by damage to the visual area of the brain [Drysdale 2009]. Object observed for a long time will suddenly appear to be expanding, shrinking, or even distorting in the patient's eyes, which may strongly influence Kollwitz's style.

According to a discussion about artistic creativity by neuroscientist Zeki, visual art follows two laws mimicking the brain's perception: to seek a constancy about the world, and to make abstraction. These laws led Picasso and Braque to eliminate perspective, distance and lighting conditions during their analytical period of Cubism [Zeki 2001]. How an artist extracts constant information and abstracts it is closely related to the artist's unique vision. The VR headset is highly suitable for simulating symptoms such as agnosia or AIWS. Moreover, its purpose is not merely for entertainment but also offers a possibility of creating or appreciating art from a pathological perspective.

8 Conclusion and Future Work

As expected, and regardless of their aptitude (or interest) to solve the cube, participants considered the experience as enjoyable and perhaps *revealing* of a new form of being in the world. After a few minutes, rather than engaging in the puzzle logic itself, many participants forgot about the gameplay and concentrated on a free exploration of the interaction dynamics, observing the eerie effects of spinning the physical faces while seeing the walls scramble around them. Some also peeked at their hands under the VR headset, either to reorient the cube axes or, perhaps, to escape the rabbit hole and the oppressive feeling of being trapped on the other side of the (looking) Rubik's Cube.

In any case, by combining a well-known 3D puzzle game and its (unusual) virtual reality representation, we may have created a simple platform for exploring questions related to the nature of physical and virtual embodiment. We hope that this simple but intriguing experiment may inspire others to explore new avenues for uses of virtual reality in games and art.

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